

Jin Hong Moon

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EDUCATION

Johns Hopkins University

Expected Dec 2026

B.S./M.S. in Computer Science, Cumulative GPA: 3.88

Baltimore, MD

Relevant Courses: Machine Learning, Deep Learning, Operating Systems, Parallel Programming, Algorithms, Databases, Linear Algebra, Probability, Statistics, System Design, Object-oriented Design

Teaching Assistant: Data Structures (Java), Intermediate Programming (C/C++)

EXPERIENCE

Machine Learning Researcher

Jan 2026 – Present

Johns Hopkins University, 3P Vision Lab

Baltimore, MD

- Designed an energy-based Vision Transformer (PEQ) that iteratively refines attention via equilibrium optimization and cross-layer feedback, improving ADE20K segmentation by 1.4 mIoU with near-zero added parameters
- Built a CUDA/C++ kernel for PEQ attention refinement using warp-shuffle and shared-memory reductions, with FP32 accumulation for FP16/BF16 inputs
- Scaled ViT training across 4 GPUs with PyTorch DistributedDataParallel and NCCL, including distributed data loading, mixed-precision training, gradient accumulation, and synchronized checkpointing

Software Engineering Intern

May 2026 – August 2026

Google

Sunnyvale, CA

- Designed and shipped a C++ caching layer eliminating redundant Gmail body-hash computations at 1.53M QPS
- Optimized cache memory access for CPU cache locality and safely reused hashes across MIME message mutations
- Validated zero cryptographic mismatches across billions of operations in production; bypassed 945K QPS, reduced hashing latency by 52.5%, and saved ~454 continuous CPU cores

Software Engineering Intern

June 2025 – August 2025

Bloomberg

New York City, NY

- Designed a distributed data pipeline for PORT<GO> time-series data layer, developing a 20+ threaded C++ producer, Python consumer, and a 4-partition Confluent Kafka topic
- Optimized read query performance of a C++ high-throughput, low-latency infrastructure processing 4B+ daily queries at 25K requests/sec, reducing average latency by 32.6%
- Built a real-time time-series analytics layer using ANSI SQL, Apache Trino, and Superset

PROJECTS

Summit - Job Application Tracker | summit-jobs.com

February 2025 – May 2025

- Created a distributed task queue using Celery and Redis in Django REST Framework to parse job-related content from Gmail via OpenAI API with 92.6% accuracy, enabling automated job tracking from users' inboxes
- Improved a lightweight DistilBERT-based classifier to filter non-job-related emails before parsing with accuracy of 96.5%, reducing token usage and improving throughput to 5K tokens/sec
- Built an OpenAI-powered chatbot with semantic search over 16K job postings by leveraging vector embeddings generated via text-embedding-3-small and stored in Supabase's pgvector, delivering personalized recommendations

Noori AI - Medical Translation | nooriai.com

February 2025 – May 2025

- Led development of a HIPAA-compliant sub-500ms medical interpreter application across 3 languages, reducing interpretation costs by over 90% for healthcare institutions serving patients with Limited English Proficiency
- Achieved 37.0 BLEU score on a MarianNMT model trained on OPUS and web-scraped medical data, preprocessed with custom tokenizers and sentence pairs alignment, under the mentorship of Dr. Philipp Koehn

TECHNICAL SKILLS

Languages: C++, Python, C, Go, Java, SQL

Frameworks: PyTorch, TensorFlow, scikit-learn, Flask, Django, Node.js

Developer Tools: CUDA, Docker, AWS, GCP, Apache Kafka

Libraries: HuggingFace, NCCL, NumPy, Pandas, Celery